

python is high level language

python easy to learn

python can do the AIML

We call ourselves Homo sapiens—man the wise—because our intelligence is so important

to us. For thousands of years, we have tried to understand how we think; that is, how a mere

handful of matter can perceive, understand, predict, and manipulate a world far larger and

more complicated than itself. The field of artificial intelligence, or AI, goes further still: it ARTIFICIAL

INTELLIGENCE

attempts not just to understand but also to build intelligent entities.

AI is one of the newest fields in science and engineering. Work started in earnest soon

after World War II, and the name itself was coined in 1956. Along with molecular biology,

AI is regularly cited as the “field I would most like to be in” by scientists in other disciplines.

A student in physics might reasonably feel that all the good ideas have already been taken by

Galileo, Newton, Einstein, and the rest. AI, on the other hand, still has openings for several

full-time Einsteins and Edisons.

AI currently encompasses a huge variety of subfields, ranging from the general (learning

and perception) to the specific, such as playing chess, proving mathematical theorems, writing

poetry, driving a car on a crowded street, and diagnosing diseases. AI is relevant to any

intellectual task; it is truly a universal field.

1.1 WHAT IS AI?

We have claimed that AI is exciting, but we have not said what it is. In Figure 1.1 we see

eight definitions of AI, laid out along two dimensions. The definitions on top are concerned

with thought processes and reasoning, whereas the ones on the bottom address behavior. The

definitions on the left measure success in terms of fidelity to human performance, whereas

RATIONALITY the ones on the right measure against an ideal performance measure, called rationality. A

system is rational if it does the “right thing,” given what it knows.

Historically, all four approaches to AI have been followed, each by different people

with different methods. A human-centered approach must be in part an empirical science, in1

2 Chapter 1. Introduction

Thinking Humanly Thinking Rationally

“The exciting new effort to make computers think ... machines with minds, in the full and literal sense.” (Haugeland, 1985)

“The study of mental faculties through the use of computational models.”

(Charniak and McDermott, 1985)

“[The automation of] activities that we associate with human thinking, activities

such as decision-making, problem solving, learning ...” (Bellman, 1978)

“The study of the computations that make it possible to perceive, reason, and act.”

(Winston, 1992)

Acting Humanly Acting Rationally

“The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil,

1990)

"Computational Intelligence is the study of the design of intelligent agents." (Poole et al., 1998)

"The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)

"AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

Figure 1.1 Some definitions of artificial intelligence, organized into four categories.

volving observations and hypotheses about human behavior. A rationalist¹ approach involves

a combination of mathematics and engineering. The various group have both disparaged and

helped each other. Let us look at the four approaches in more detail.

1.1.1 Acting humanly: The Turing Test approach

TURING TEST The Turing Test, proposed by Alan Turing (1950), was designed to provide a satisfactory

operational definition of intelligence. A computer passes the test if a human interrogator, after

posing some written questions, cannot tell whether the written responses come from a person

or from a computer. Chapter 26 discusses the details of the test and whether a computer would

really be intelligent if it passed. For now, we note that programming a computer to pass a

rigorously applied test provides plenty to work on. The computer would need to possess the

following capabilities:

- natural language processing to enable it to communicate successfully in English; NATURAL LANGUAGE PROCESSING

- knowledge representation to store what it knows or hears; KNOWLEDGE REPRESENTATION

AUTOMATED

REASONING • automated reasoning to use the stored information to answer questions and to draw

new conclusions;

MACHINE LEARNING • machine learning to adapt to new circumstances and to detect and extrapolate patterns.

¹ By distinguishing between human and rational behavior, we are not suggesting that humans are necessarily

"irrational" in the sense of "emotionally unstable" or "insane." One merely need note that we are not perfect:

not all chess players are grandmasters; and, unfortunately, not everyone gets an A on the exam. Some systematic

errors in human reasoning are cataloged by Kahneman et al. (1982)